AYNA MUSAYEVA

INVESTIGATIVE RESEARCHER / SOFTWARE DEVELOPER

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PROFESSIONAL PROFILE

Investigative researcher and software developer specializing in corporate intelligence, financial crime research, OSINT, and evidence-based analysis. Builds Python and SQL investigative applications spanning corporate networks, financial transactions, authentication activity, and directed transaction graphs.

CORE COMPETENCIES

INVESTIGATIONS Corporate intelligence • OSINT • Due diligence • Securities filings • Regulatory analysis • Source verification
ENGINEERING & DATA Python • SQL • Linux • Bash • Pandas • NumPy • MySQL • DuckDB • Streamlit • Git • GitHub • Pytest
ANALYTICS & AI Graph analytics • Entity resolution • Data validation • RAG • FAISS • Vector search • NLP • OCR • AI evaluation

PROFESSIONAL EXPERIENCE

Financial & Regulatory Investigative Researcher Nov 2024–Present
Capital, Ownership, and Opacity

- Investigate cross-border ownership, corporate relationships, financing structures, and regulatory oversight through filings, securities disclosures, legal records, and multilingual OSINT.
- Reconstruct entity networks, reconcile conflicting sources, and document evidence-backed findings while distinguishing verified facts from inference.

AI Research Infrastructure Developer Jan 2025–Present
Nimillux

- Designed a local Python research system integrating hybrid retrieval, source-grounded RAG, evidence provenance, and citation verification across heterogeneous document collections.
- Implemented claim-support checks, confidence assessment, and regression tests to identify unsupported outputs and evaluate system limitations.

Marketing Specialist Aug 2023–Apr 2025
APOGEE Media

- Built automated customer journeys and targeted campaigns using Salesforce Marketing Cloud, Journey Builder, Automation Studio, and Email Studio.
- Applied SQL, audience segmentation, and campaign-performance analysis to support CRM workflows and digital marketing decisions.

Editor 2018–2019
Musée Magazine

- Researched and fact-checked print and digital features, evaluated sources, and coordinated contributor recruitment and editorial workflows.

SELECTED ENGINEERING PROJECTS

CRIE | Corporate Relationship Intelligence Engine Built Python/MySQL/Streamlit workflows for GLEIF corporate networks, evidence reconciliation, and verifiable reports; 782 database-free tests passed.

TRIS | Transaction Risk Intelligence System Processed 6.36M synthetic financial transactions through explainable monitoring rules, transaction networks, case management, and held-out evaluation.

ATIS | Account Takeover Intelligence System Analyzed 31.27M synthesized authentication events with anomaly investigations, account/IP relationships, traceable evidence, and case audits.

BTGI | Bitcoin Transaction Graph Investigator Analyzed 203,769 anonymized transaction nodes and 234,355 directed edges using bounded graph traversal and evidence-linked structural analysis.

EDUCATION & CERTIFICATIONS

Harvard | Certificate in Data Science and AI Principles • 2025–2026
Western University | B.A., Business Administration • 2012–2016
International Center of Photography | Documentary Practices & Visual Journalism • 2020–2021
Languages English • Russian • Turkish